آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

# Notes from E-Marking Centre SSC-I General Mathematics Annual Examinations 2024

## Introduction

This document has been prepared for the teachers and candidates of Secondary School Certificate (SSC) Part I (Class IX) General Mathematics. It contains comments on candidates' responses to the 2024 SSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

## E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

## General Observations

Candidates performed really well in some concepts, such as, Sets, Matrices and Ratio. However, candidates who did not score well mostly failed to understand the demands of the questions, often misinterpreting the command words and the stimuli.

Mentioned below are few concepts that teachers need to focus so that the candidates may perform better.

- Direct and Inverse Variations
- Business Partnership
- Algebraic Identities
- Factorisation
- Linear Graphs
- Logarithm

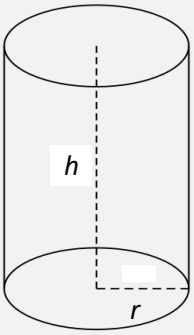
**Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.**

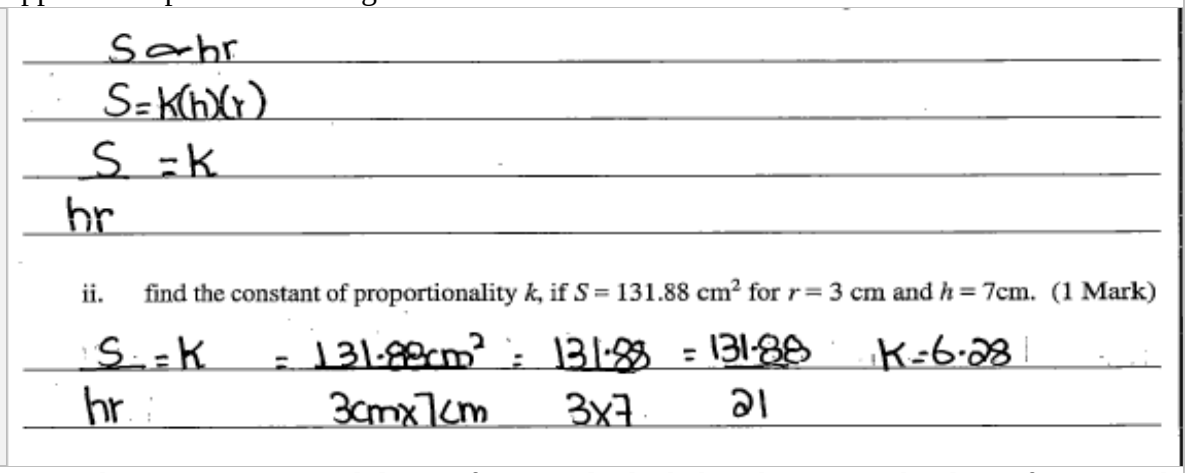
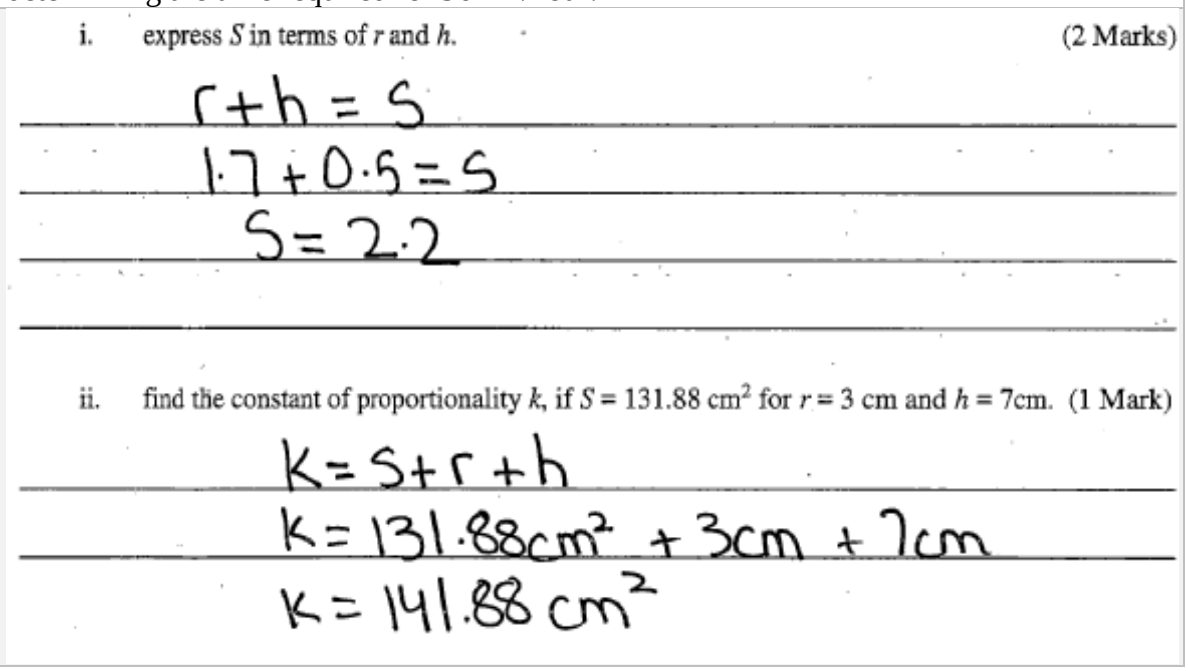
## DETAILED COMMENTS

### Constructed Response Questions (CRQs)

**Question No. 1a**

Candidates were given the choice to attempt any ONE out of the two questions: 1a and 1b. Majority of the candidates attempted 1b.

| | |
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| **Question Text** | The surface area of a cylinder is directly proportional to its height '$h$' and radius '$r$'<br>  <br><br>If the surface area is represented by $S$, then<br>i. express $S$ in terms of $r$ and $h$<br>ii. find the constant of proportionality $k$, if $S$ = 131.88 cm$^2$ for $r$ = 3cm and $h$ = 7cm. |
| **SLO No.** | 1.4.3 |
| **SLO Text** | Solve problems involving direct, inverse, compound proportion and variation ($x$ varies directly or inversely as $y$). |
| **Max Marks** | 3 |
| **Cognitive Level** | A* |
| **Checking Hints** | i. 1 mark for combining the direct proportions and writing in the form $S \propto rh$<br>1 mark for expressing the proportionality in form of an equation as $S = k\ rh$<br>ii. 1 mark for finding the value of $k$ |
| **Overall Performance** | The question assessed candidates' understanding of direct proportion. Many candidates struggled to differentiate between direct and inverse proportion and to solve related equations. Few candidates correctly identified and applied direct proportion with the provided quantities. |

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| **Description of Better Responses** | In this task, candidates were required to solve the problem in accordance with STEM standards, employing correct mathematical notation and adhering to rules and regulations. Better responses showed that candidates have completed the solution step-by-step, ensuring that numerical values are substituted accurately as per the given guidelines. The final answers met the specified requirements and demonstrated a clear, methodical approach to problem-solving. |
| **Image of Better Response** |  |
| **Description of Weaker Responses** | In weaker responses, candidates often overlooked the phrase "12 km/hour faster" and failed to incorporate the given speed of 84 km/hour when calculating for 96 km/hour. As a result, many of them incorrectly computed the time for 12 km/hour instead of correctly determining the time required for 96 km/hour. |
| **Image of Weaker Response** |  |

**Suggestions for improvement (Highlight all that apply)**

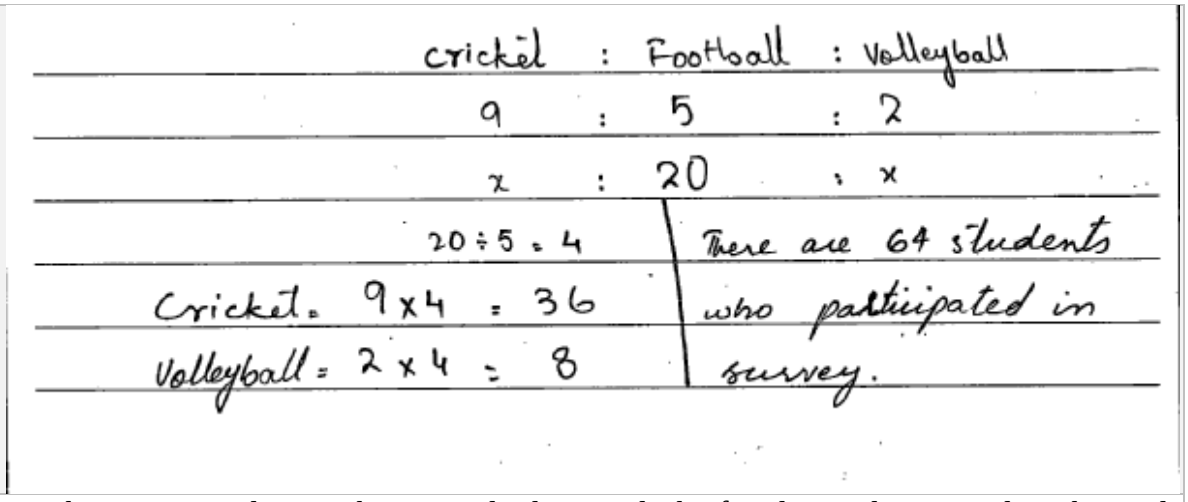
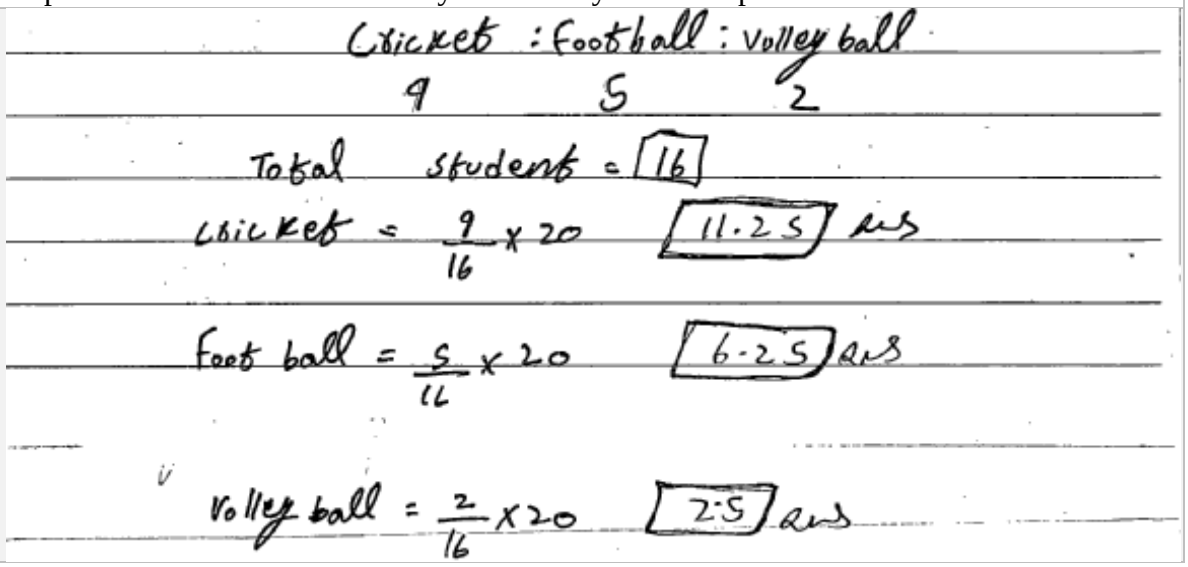
| Maximising SLO Achievement | Preferred Pedagogy** Used for this SLO | Assessment Strategies |
|---|---|---|
| • Identify the expectation of command words (use Command Word Guide)<br>• Ensure the content is taught at the relevant cognitive level<br>• Identify necessary content required (skills + | • Story Board<br>• Cause and Effect<br>• Fish and Bone<br>• Concept Mapping<br>• Audio Visual Resources<br>• Think, Pair and Share<br>• Knowledge Platform | • Past paper questions<br>• Discussion on E-Marking Notes<br>• AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login |

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| concepts)<br>• Review past paper questions on the concept<br>• Utilise the resource guide for additional materials | videos<br>• Questioning Technique (Socratic approach)<br>• Practical Demonstration<br><br>** For description of each Pedagogy, refer to Annexure A | |

**Any Additional Suggestion:**
To help with the misunderstandings seen in responses, consider offering extra learning materials that highlight the special features of different proportional and variational situations. Focus on explaining the differences between direct and inverse variations in everyday situations to clear up the confusions.

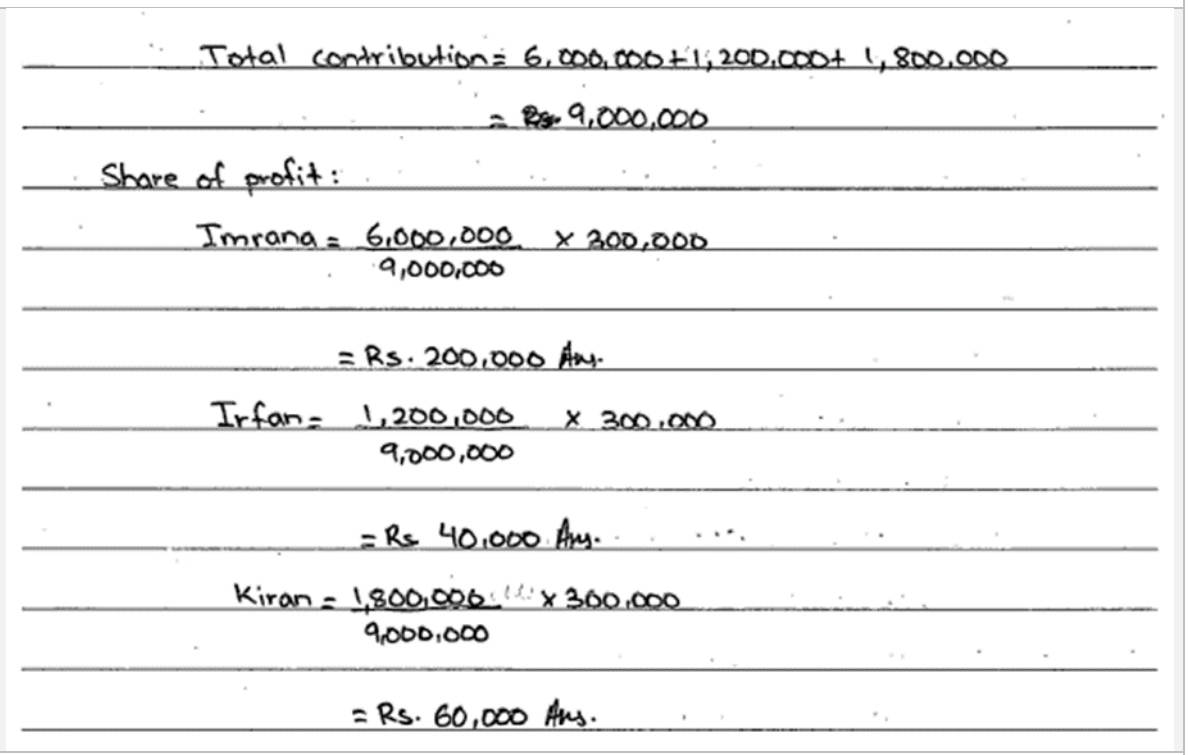
*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

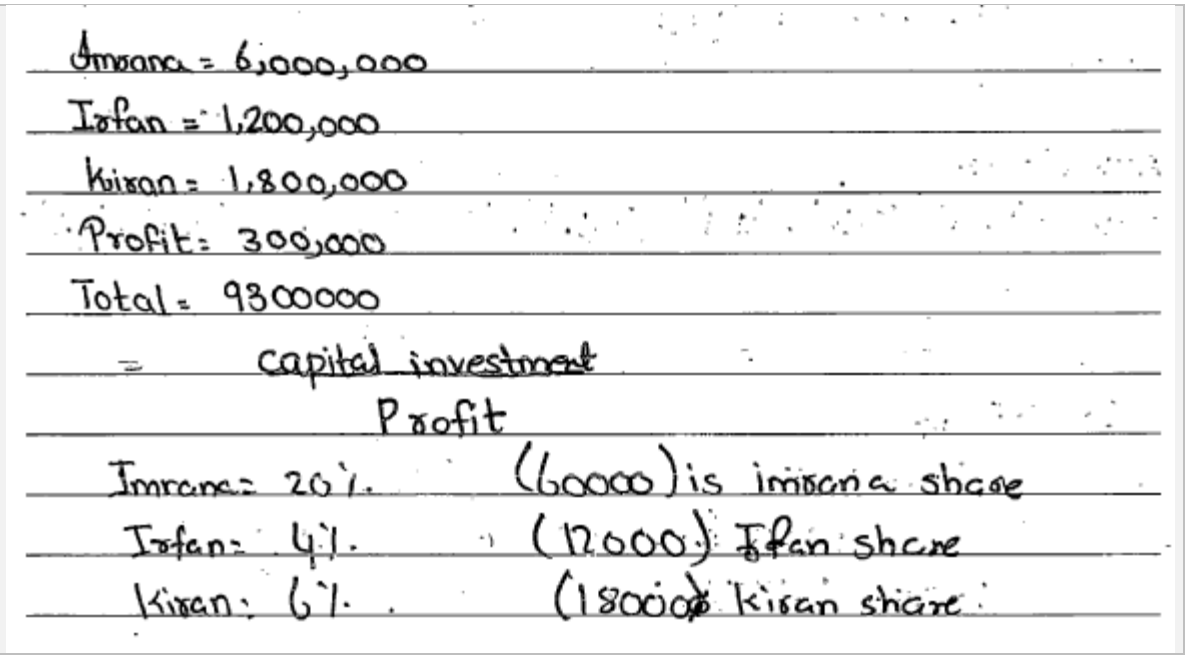
| Question No. 1b | |
|---|---|
| **Question Text** | A sports teacher conducted a survey to inquire about the game choices of students in a school. The available choices were cricket, football and volleyball. It was found that the ratio of students who chose cricket, football and volleyball was 9:5:2 respectively.<br>If the number of students choosing football is 20, then find the total number of students who participated in the survey. |
| **SLO No.** | 1.3.3 |
| **SLO Text** | Solve word problems involving ratio; |
| **Max Marks** | 3 |
| **Cognitive Level** | A |
| **Checking Hints** | 1 mark for finding the sum of ratio (indication of selecting the method)<br><br>1 mark for constructing the relation as $20 = \frac{5}{16} \times x$<br><br>1 mark for calculating the value of $x$ |
| **Overall Performance** | This question was related to the topic of ratio and proportion. Majority of the candidates attempted this question correctly and obtained full marks. Some of the candidates struggled to use ratios for finding the quantity which reflects the need for further clarity in the topic. |
| **Description of Better Responses** | Better responses demonstrated clear understanding of solving problems related to ratios. The responses depicted constructing the equation and allotting the variable to the unknown quantity with respect to the given situation. Candidates constructed this relation $20 = \frac{5}{16} \times x$ to solve the given problem and found the value of x. |

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|---|---|
| **Image of Better Response** |  |
| **Description of Weaker Responses** | Weaker responses lost marks primarily due to a lack of understanding regarding the total number or the concept of sharing parts in ratio-related questions. They found the sum of ratios as 14 or 15 at the place of 16. This gap in knowledge led to inaccuracies in their responses and affected their ability to correctly solve the problem. |
| **Image of Weaker Response** |  |

**Suggestions for improvement (Highlight all that apply)**

| Maximising SLO Achievement | Preferred Pedagogy Used for this SLO | Assessment Strategies |
|---|---|---|
| • Identify the expectation of command words (use Command Word Guide)<br>• Ensure the content is taught at the relevant cognitive level<br>• Identify necessary content required (skills + concepts)<br>• Review past paper questions on the concept<br>• Utilise the resource guide for additional materials | • Story Board<br>• Cause and Effect<br>• Fish and Bone<br>• Concept Mapping<br>• Audio Visual Resources<br>• Think, Pair and Share<br>• Knowledge Platform videos<br>• Questioning Technique (Socratic approach)<br>Practical Demonstration | • Past paper questions<br>• Discussion on E-Marking Notes<br>• AKU-EB Digital Learning Solution powered by Knowledge Platform<br>https://akueb.knowledgeplatform.com/login |
| **Any Additional Suggestion:** | | |

| Question No. 2 | |
|---|---|
| **Question Text** | Imrana started a business with a capital of Rs 6,000,000. Irfan joined the business after 6 months contributing a capital of Rs 1,200,000. After three months of Irfan's joining, Kiran joined the business with a capital of Rs 1,800,000. At the end of the year, the business earned a profit of Rs 300,000.<br><br>Calculate the share of each person in the profit.<br><br>(**Note:** Assuming all the three persons will get the profit for whole year.) |
| **SLO No.** | 3.3.2 |
| **SLO Text** | Solve word problems based on business partnership (at most four partners). |
| **Max Marks** | 4 |
| **Cognitive Level** | A |
| **Checking Hints** | 1 mark for taking the correct ratio and writing in simplest form<br><br>1 mark for finding the required share in the profit (3 required) |
| **Overall Performance** | This question pertained to the topic of business partnerships and candidates approached it using various methods. The candidates not only understood the context of the question but also paid attention to the details and solved it using the appropriate method. |
| **Description of Better Responses** | Better responses applied the appropriate method by first establishing ratios between investments and then determining each person's share of the profit based on these ratios. This methodical approach ensured they correctly addressed the partnership question and accurately calculated the distribution of profits according to each partner's investment contribution. |
| **Image of Better Response** |  |
| **Description of Weaker Responses** | The common error identified was that some cadidates struggled to distinguish between investment and profit in the given problem. Additionally, some candidates made errors in calculating ratios by incorrectly adding the investments together and treating the fractional part of their investment as profit. |

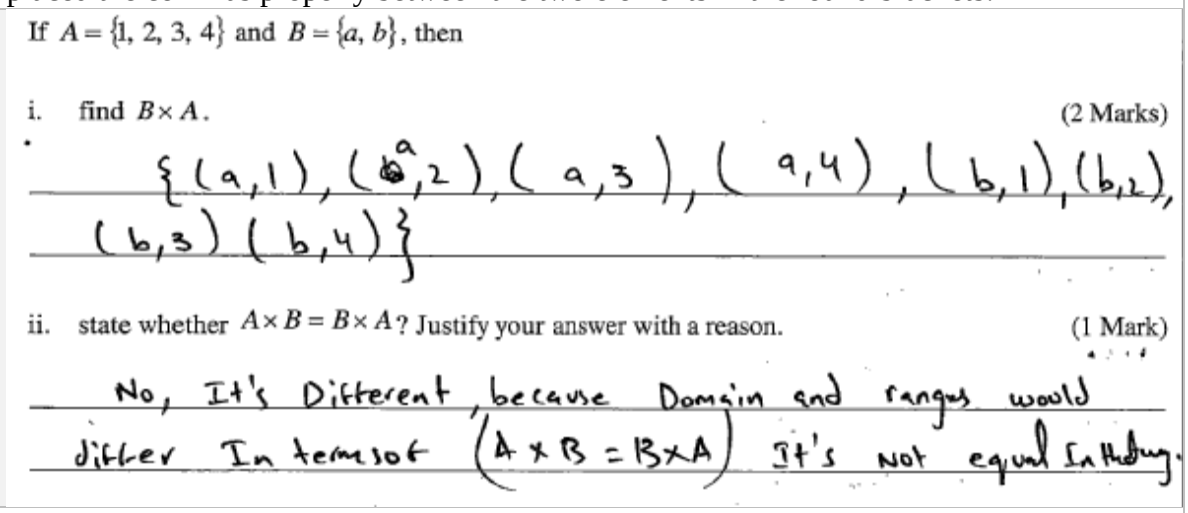
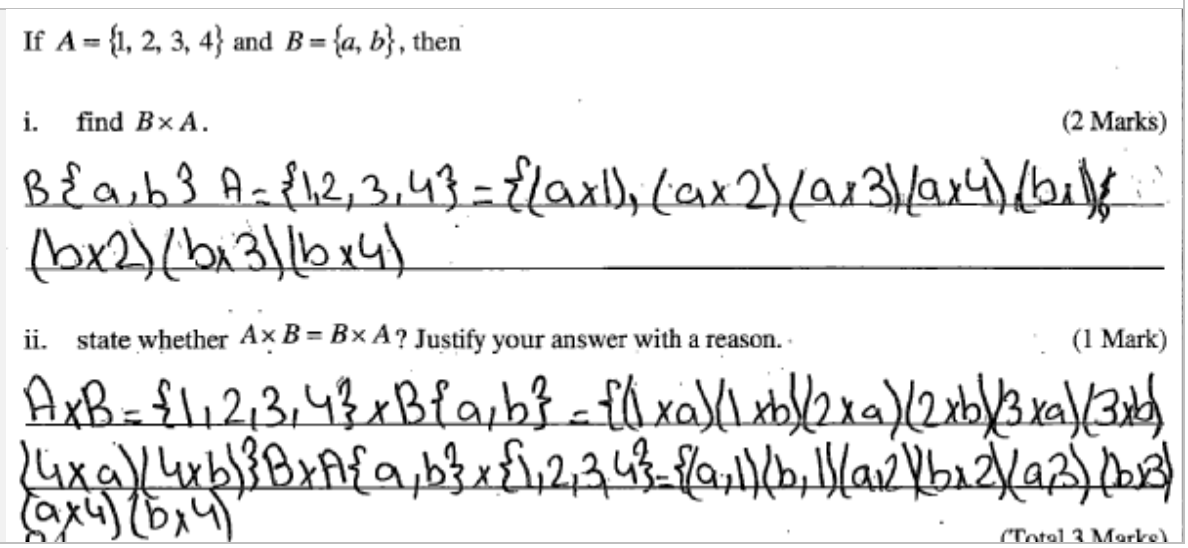
| | |
|---|---|
| **Image of Weaker Response** |  |

**Suggestions for improvement (Highlight all that apply)**

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|---|---|---|
| • Identify the expectation of command words (use Command Word Guide)<br>• Ensure the content is taught at the relevant cognitive level<br>• Identify necessary content required (skills + concepts)<br>• Review past paper questions on the concept<br>• Utilise the resource guide for additional materials | • Story Board<br>• Cause and Effect<br>• Fish and Bone<br>• Concept Mapping<br>• Audio Visual Resources<br>• Think, Pair and Share<br>• Knowledge Platform videos<br>• Questioning Technique (Socratic approach)<br>• Practical Demonstration | • Past paper questions<br>• Discussion on E-Marking Notes<br>• AKU-EB Digital Learning Solution powered by Knowledge Platform<br>https://akueb.knowledgeplatform.com/login |

**Any Additional Suggestion:** Teachers may use real life examples to calculate the share of each individual profits.

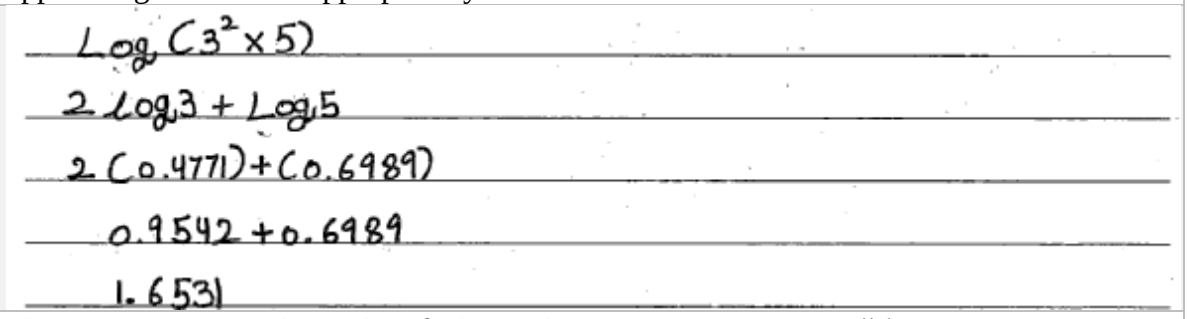
| Question No. 3 | |
|---|---|
| **Question Text** | If $A = \{1, 2, 3, 4\}$ and $B = \{a, b\}$, then<br>i. find $B \times A$.<br>ii. state whether $A \times B = B \times A$? Justify your answer with a reason. |
| **SLO No.** | 4.4.2 |
| **SLO Text** | Solve problems related to SLO 4.4.1; |
| **Max Marks** | 3 |
| **Cognitive Level** | U |
| **Checking Hints** | i. 1 mark for four correct ordered pairs (2 required)<br>ii. 1 mark for stating the condition for $A \times B = B \times A$ |

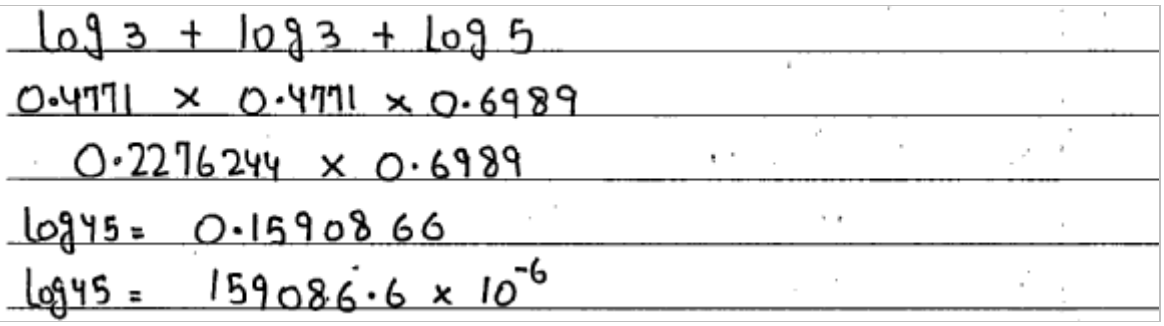
| | |
|---|---|
| **Overall Performance** | This question was related to understanding of the domain and range concept. Candidates were required to correctly verify $A \times B = B \times A$. Less than half of the candidates attempted this question correctly and obtained the desired results. |
| **Description of Better Responses** | Candidates demonstrated understanding of domain and range concept through making correct ordered pairs, using brackets as required. They also grasped the concept of proving C. These responses correctly selected the appropriate brackets for their positions and placed the commas properly between the two elements in the round brackets. |
| **Image of Better Response** |  If $A = \{1, 2, 3, 4\}$ and $B = \{a, b\}$, then<br>i. find $B \times A$. (2 Marks)<br>{(a,1), (b,2), (a,3), (a,4), (b,1), (b,2), (b,3) (b,4)}<br>ii. state whether $A \times B = B \times A$? Justify your answer with a reason. (1 Mark)<br>No, It's Different, because Domain and ranges would differ In termsof $(A \times B = B \times A)$ It's Not equal In Nature. |
| **Description of Weaker Responses** | Candidates lacked understanding of ordered pairs and the Cartesian product in sets, which are fundamental concepts in binary relations. They also struggled to apply appropriate reasoning when discussing these concepts. In (i) some of them solved for $A \times B$ instead of $B \times A$ which changed the position of the elements and solution went wrong. |
| **Image of Weaker Response** |  If $A = \{1, 2, 3, 4\}$ and $B = \{a, b\}$, then<br>i. find $B \times A$. (2 Marks)<br>B{a,b} A={1,2,3,4} ={(ax1), (ax2)(ax3)(ax4)(bx1)(bx2)(bx3)(bx4)<br>ii. state whether $A \times B = B \times A$? Justify your answer with a reason. (1 Mark)<br>AxB={1,2,3,4}xB{a,b} ={(1xa)(1xb)(2xa)(2xb)(3xa)(3xb)(4xa)(4xb)}BxA{a,b}x{1,2,3,4}={(a,1)(b,1)(a,2)(bx2)(a,3)(b,3)(ax4)(bx4)<br>(Total 3 Marks) |

**Suggestions for improvement (Highlight all that apply)**

| Maximising SLO Achievement | Pedagogy Used for that SLO | Assessment Strategies |
|---|---|---|
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|---|---|---|
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| **Any Additional Suggestion:** | | |

| **Question No. 4** | |
|---|---|
| **Question Text** | If $\log 3 = 0.4771$ and $\log 5 = 0.6989$, then find the value of $\log 45$. |
| **SLO No.** | 5.5.1 |
| **SLO Text** | Apply the following laws of logarithm to solve related problems (without using log and antilog tables). |
| **Max Marks** | 3 |
| **Cognitive Level** | A |
| **Checking Hints** | 1 mark for writing sum of logs of 9 and 5<br>1 mark for applying the third law of logarithm and writing $= 2\log 3 + \log 5$<br>1 mark for adding the logs and finding the value |
| **Overall Performance** | In this question, which involves the application of logarithm laws, overall candidates' performance remains below average. However, those candidates who had clear understanding of the logic behind these logarithm laws performed well. |
| **Description of Better Responses** | Better responses demonstrated the strong understanding of the application of logarithm laws. They correctly showed addition and multiplication where needed. These candidates accurately identified the requirements of the question, identified relevant factors and applied logarithm laws appropriately to find the desired value. |
| **Image of Better Response** |  |
| **Description of Weaker Responses** | The common mistakes identified in this question was candidates' misconception regarding the application of logarithm laws. Many candidates faced difficulty in finding the factors of 45 and some candidates directly substituted the given values into their calculations, resulting in deductions of marks. |

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| **Image of Weaker Response** |  |

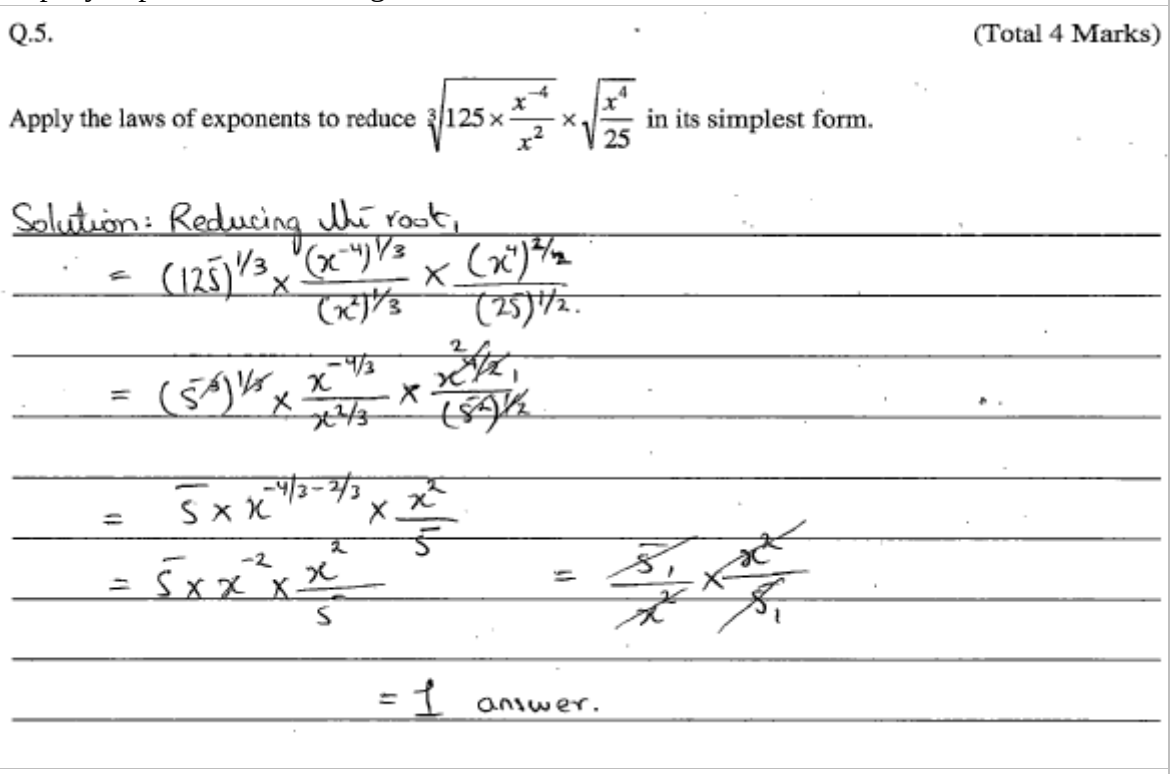
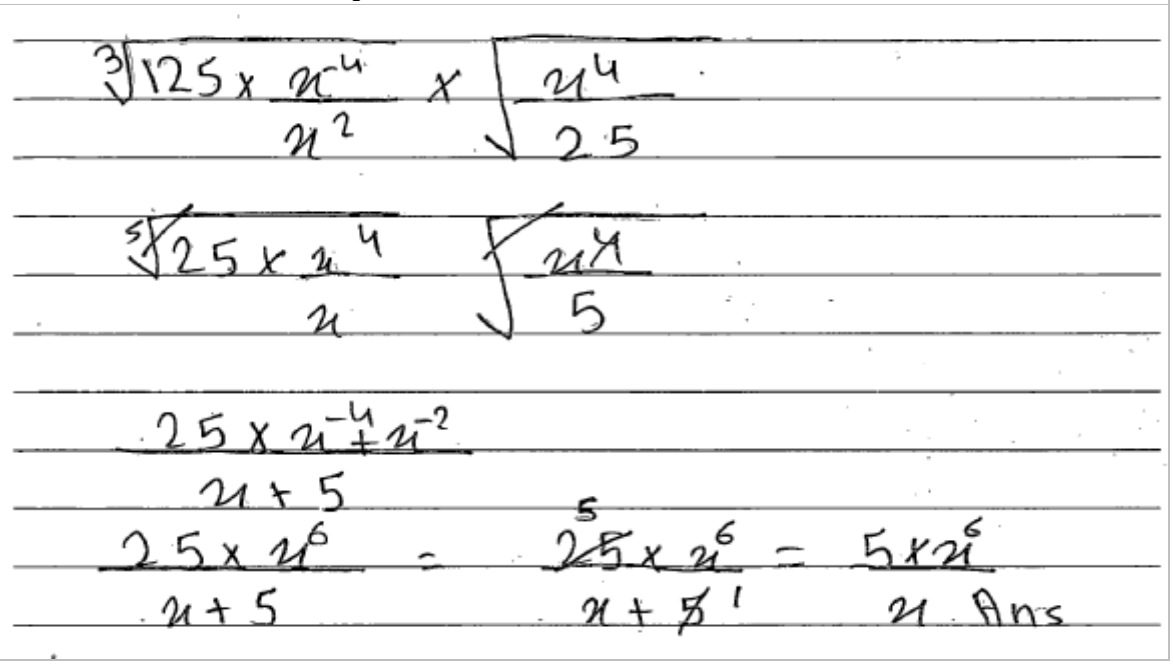
**Suggestions for improvement (Highlight all that apply)**

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**Any Additional Suggestion:**
While teaching the concept of application of the laws of logarithm, teachers are advised to use the following strategies:

1. Begin by providing a clear and concise explanation of each logarithmic law.
2. Provide students with incorrect solutions and ask them to identify and explain the errors.

| **Question No. 5** | |
|---|---|
| **Question Text** | Apply the laws of exponents to reduce $\sqrt[3]{125\times\frac{x^{-4}}{x^2}}\times\sqrt{\frac{x^4}{25}}$ in its simplest form. |
| **SLO No.** | 6.1.5 |
| **SLO Text** | Solve problems based on multiplication and division of rational expressions. |
| **Max Marks** | 4 |
| **Cognitive Level** | A |
| **Checking Hints** | 1 mark for writing negative (-ve) exponents as positive exponents<br>1 mark for writing radical to exponent form<br>1 mark for writing $125\times\frac{1}{x^6}=5^3\times\left(\frac{1}{x^2}\right)^3$ and $\frac{x^4}{25}=\left(\frac{x^2}{5}\right)^2$<br>1 mark for simplification |
| **Overall Performance** | This question was related to the application of laws of exponents to simplify the given expression. Only few of the candidates were able to meet the demand of the question. Majority of the candidates struggled in simplifying the exponents and taking the square roots and cube roots. |

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| **Description of Better Responses** | Stronger responses showcased a solid understanding of prime factorization and the application of index rules. Candidates effectively utilized square roots and cube roots, demonstrating their ability to simplify exponents accurately. They not only identified the factors correctly but also applied the relevant mathematical principles to manipulate and simplify expressions, resulting in clear and concise solutions. |
| **Image of Better Response** |  |
| **Description of Weaker Responses** | Canidates struggled in this question due to lack of understanding of surds, indices and exponents. They also made mistakes in applying square roots, cube roots, addition and subtraction correctly and understanding the specific requirements of the question. This resulted in errors and incomplete solutions. |
| **Image of Weaker Response** |  |

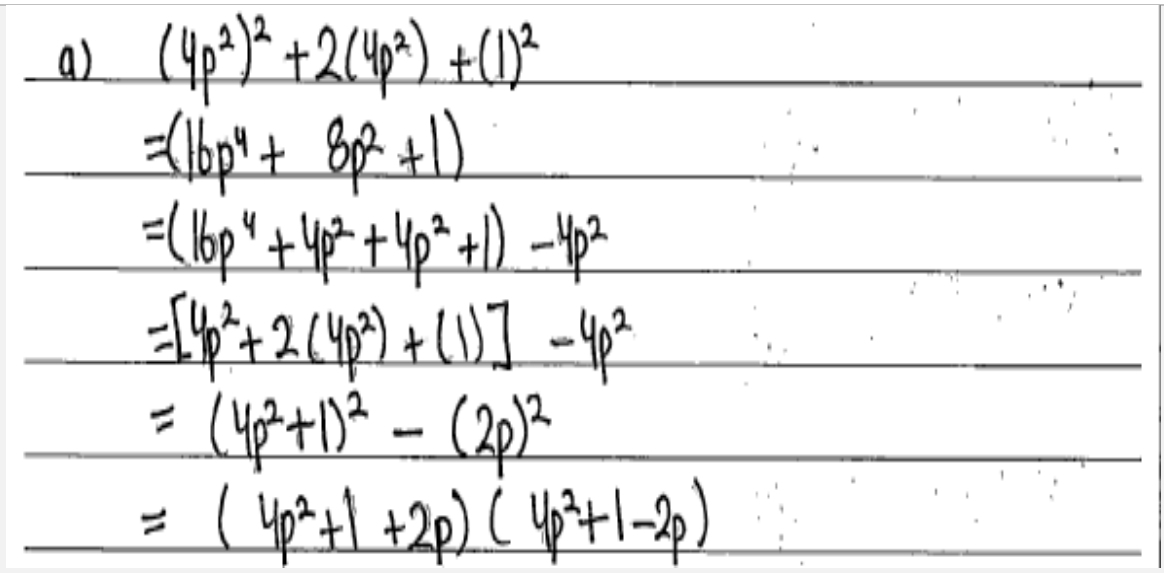
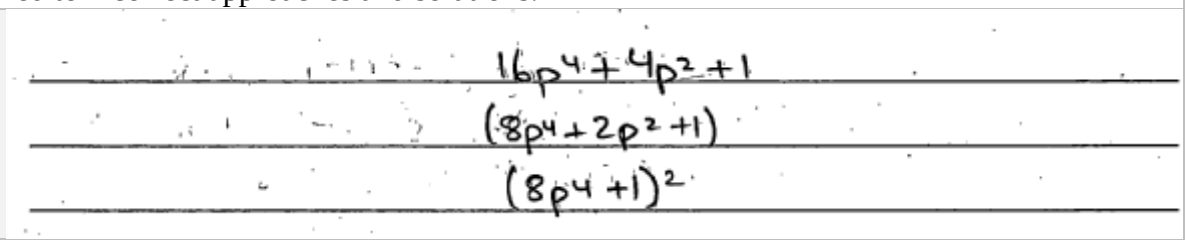
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| **Any Additional Suggestion:** | | |

**Question No. 6a**

Candidates were given the choice to attempt any ONE out of the two questions: 6a and 6b. Majority of the candidates attempted 6a.

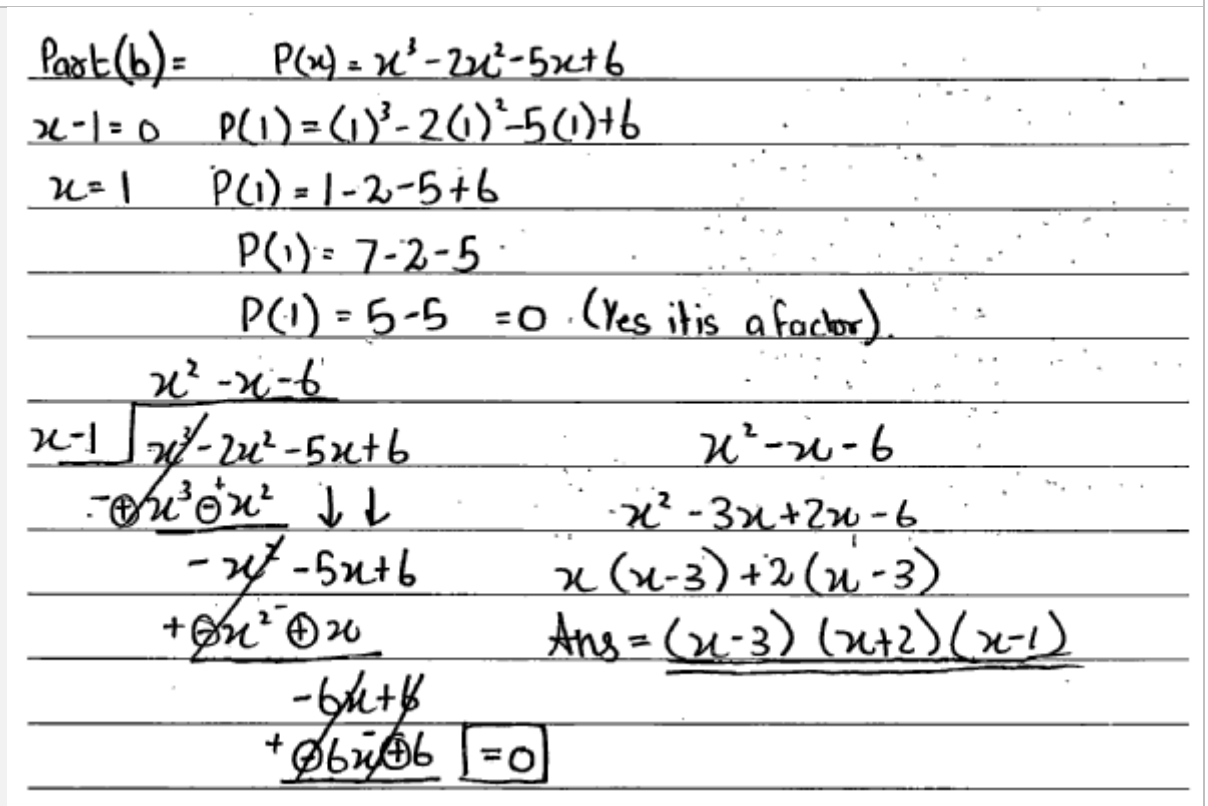
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|---|---|
| **Question Text** | Factorise the expression $16p^4 + 4p^2 + 1$ completely. |
| **SLO No.** | 7.1.1f |
| **SLO Text** | Factorise the expression of the following types<br>$a^4 + a^2 b^2 + b^4$ or $a^4 + 4b^4$ |
| **Max Marks** | 4 |
| **Cognitive Level** | A |
| **Checking Hints** | 1 mark for writing the squares in the form $(4p^2)^2 + 4p^2 + (1)^2$<br>1 mark for identifying the right terms to be added and subtracted as $\pm 4p^2$<br>1 mark for writing in the form $= (4p^2 + 1)^2 - 4p^2$<br>1 mark for writing $= (4p^2 + 1 + 2p)(4p^2 + 1 - 2p)$ |
| **Overall Performance** | In this choice question, very few candidates achieved desired results. Those who successfully applied the relevant method of factorisation were able to attain full marks in their responses as the demand of the question was to factorise by completing square and use appropriate identities. |
| **Description of Better Responses** | Better responses correctly identified that the given expression is not a perfect square. they were able to transform it into a perfect square by adding and $\pm 4p^2$. They then proceeded to fully factorise the expression and wrote this form $(4p^2 + 1)^2 - 4p^2$, demonstrating a clear understanding of method and achieving correct factors $(4p^2 + 1 + 2p)(4p^2 + 1 - 2p)$. |

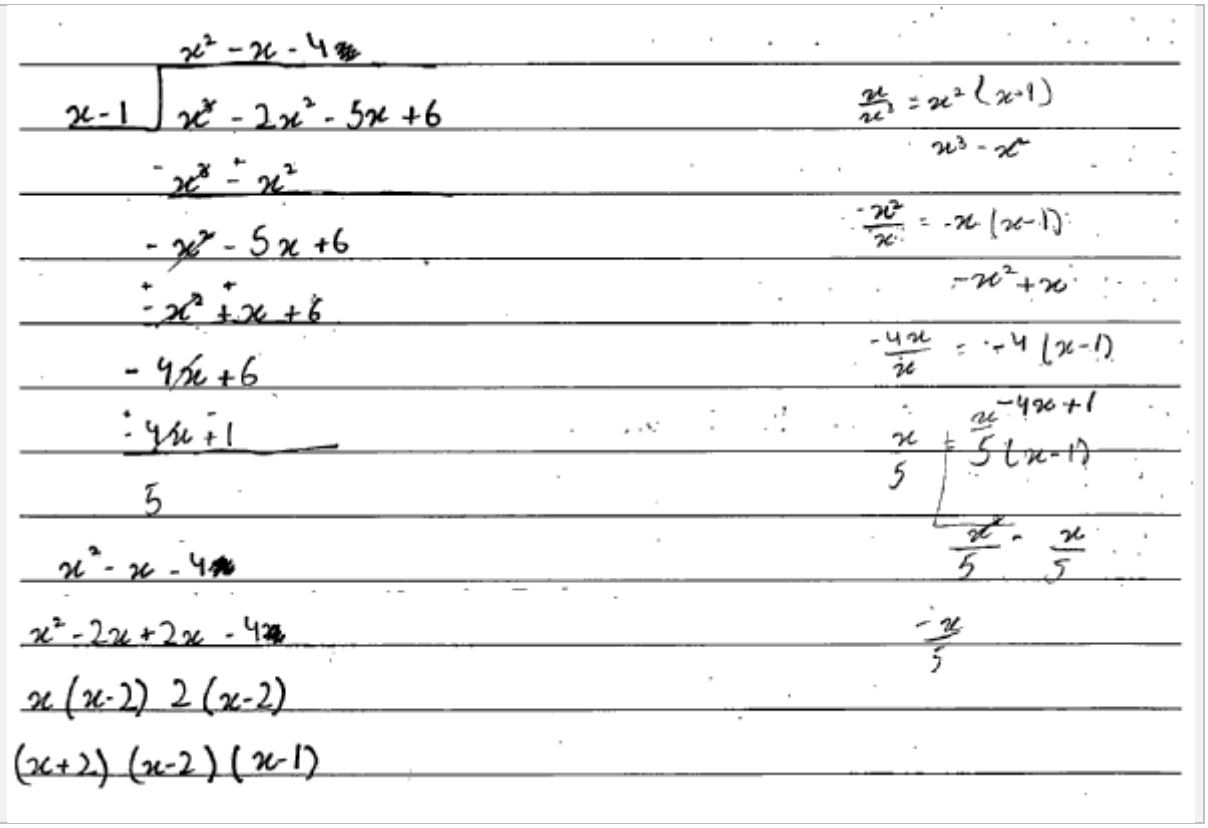
| | |
|---|---|
| **Image of Better Response** |  |
| **Description of Weaker Responses** | Weaker responses mistakenly assumed the expression to be a perfect square due to the presence of even powers. Consequently, they applied the square identity without first verifying its applicability. Some candidates attempted to use the middle term breaking method, which was also not suitable for the given expression. These misunderstandings led to incorrect approaches and solutions. |
| **Image of Weaker Response** |  |

**Suggestions for improvement (Highlight all that apply)**

| Maximising SLO Achievement | Pedagogy Used for that SLO | Assessment Strategies |
|---|---|---|
| • Identify the expectation of command words (use Command Word Guide)<br>• Ensure the content is taught at the relevant cognitive level<br>• Identify necessary content required (skills + concepts)<br>• Review past paper questions on the concept<br>• Utilise the resource guide for additional materials | • Story Board<br>• Cause and Effect<br>• Fish and Bone<br>• Concept Mapping<br>• Audio Visual Resources<br>• Think, Pair and Share<br>• Knowledge Platform videos<br>• Questioning Technique (Socratic approach)<br>• Practical Demonstration | • Past paper questions<br>• Discussion on E-Marking Notes<br>• AKU-EB Digital Learning Solution powered by Knowledge Platform<br>https://akueb.knowledgeplatform.com/login |

**Any Additional Suggestion:**
Teachers are encouraged to emphasize the distinction between like and unlike terms. They should provide exercises that focus on accurately identifying and grouping these terms. Additionally, engaging students in activities that involve adding terms while applying this concept can help improve their accuracy in mathematical operations.

| Question No. 6b | |
|---|---|
| **Question Text** | Using the factor theorem,<br>i. show that $x-1$ is a factor of $x^3-2x^2-5x+6$.<br>ii. find the remaining factors of $x^3-2x^2-5x+6$. |
| **SLO No.** | 7.3.1 |
| **SLO Text** | Apply factor theorem to factorise a cubic polynomial. |
| **Max Marks** | 4 |
| **Cognitive Level** | A |
| **Checking Hints** | i. 1 mark for writing the factor theorem correctly<br>1 mark for showing that $x-1$ is a factor of $x^3-2x^2-5x+6$<br>ii. 1 mark for substituting $-2$ or $3$ in the factor theorem<br>1 mark for identifying the factor correctly as $(x+2)$ or $(x-3)$ |
| **Overall Performance** | This question was about the factor theorem, and those who tackled it carefully earned full marks. Some candidates used the division method (which was not required), while others chose the factor method. In both cases, candidates found all three necessary factors. |
| **Description of Better Responses** | In better responses, candidates first verified the given factor. Then, for finding the remaining factors, they applied either the division method or the factor method accurately and they got the factors $-2$ or $3$. As a result, they achieved desired scores. |
| **Image of Better Responses** |  |
| **Description of Weaker Responses** | According to the question's requirements, candidates were expected to verify one of the given factors and determine the remaining two. However, it appears that many candidates misunderstood the question. Some made errors during simplification, while others failed to find the remaining factors when the given factor was not verified. |

| | |
|---|---|
| **Image of Weaker Response** |  |



**Suggestions for improvement (Highlight all that apply)**

| Maximising SLO Achievement | Preferred Pedagogy Used for this SLO | Assessment Strategies |
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**Any Additional Suggestion:**
Offer a range of problems that require the factor theorem for factorisation. Provide both straightforward and complex examples to foster confidence and adaptability.

**Question No.7**

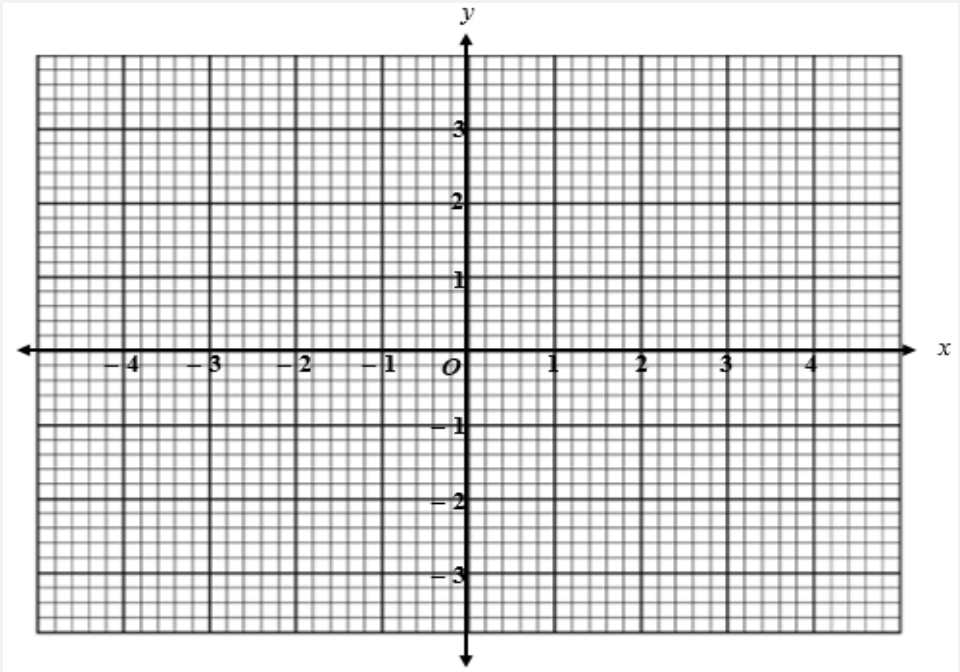
**Question Text**

A straight line is represented by the equation $y = \frac{1}{2}x + 1$ . Using this equation,

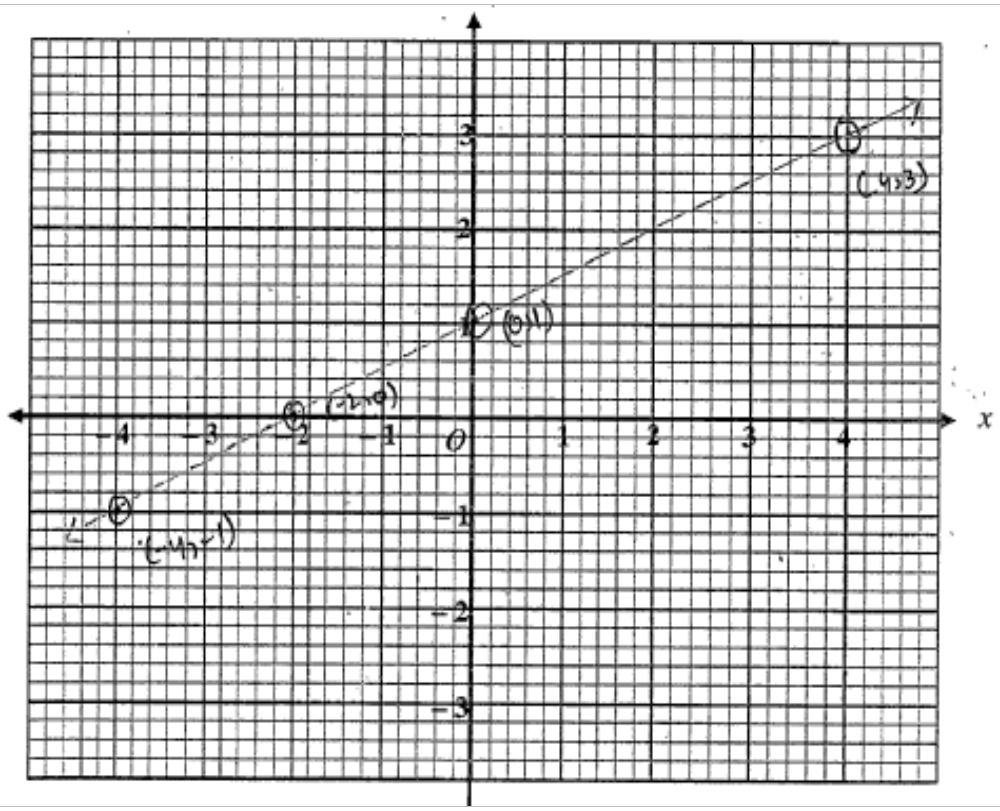
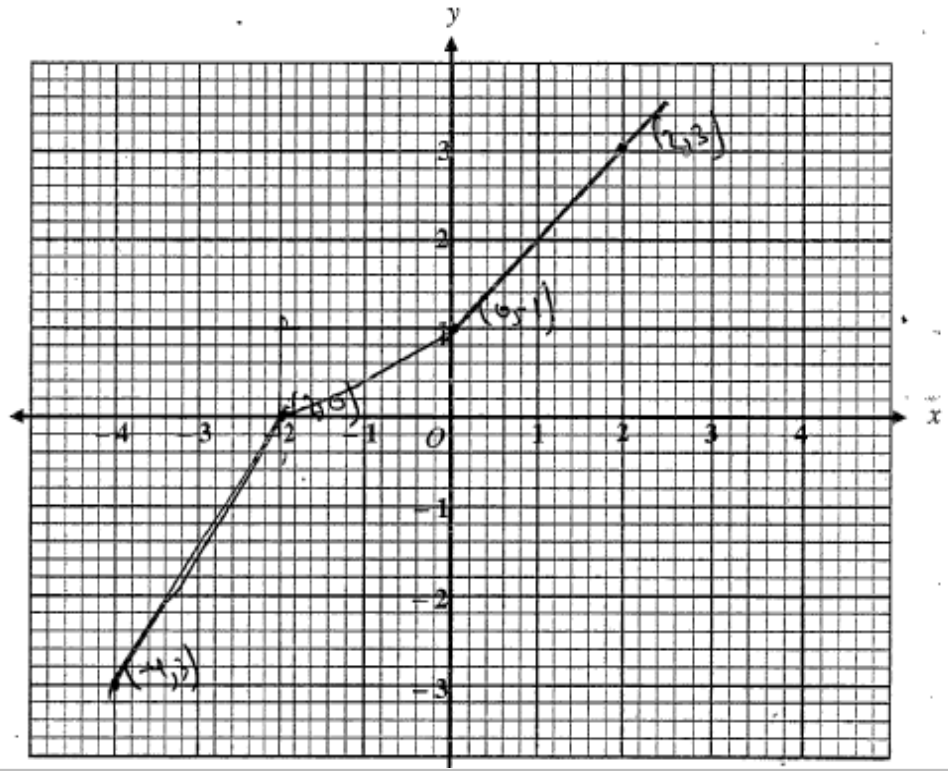
i. Complete the given table.

| $x$ | $-4$ | $-2$ | 0 | |
|---|---|---|---|---|
| $y = \frac{1}{2}x + 1$ | | 0 | 1 | 3 |

ii. Using the given table, draw the graph of the graph for $y = \frac{1}{2}x + 1$.




| | |
|---|---|
| **SLO No.** | 8.1.6 |
| **SLO Text** | Draw the graph of given linear equations d) $y = mx + c$. |
| **Max Marks** | 3 |
| **Cognitive Level** | A |
| **Checking Hints** | i. 1 mark for finding both missing values<br>ii. 1 mark for plotting at least two points correctly<br>1 mark for connecting the points and drawing the graph |
| **Overall Performance** | This question focused on linear equations and assessed two key concepts: solving a linear equation and accurately representing a straight line graphically. Most candidates could complete the table, a significant majority struggled with accurately plotting points on the graph. |
| **Description of Better Responses** | Better responses exhibited precise and accurate graphical representations. Candidates skillfully plotted points with great accuracy and drew a straight line that intersected all of those points seamlessly. Additionally, they demonstrated the ability to find the correct values of \(x\) and \(y\) as required, ensuring that their calculations were consistent with the graphical data. This attention to detail highlighted their understanding of the relationship between the variables and their ability to convey this visually. |

| | |
|---|---|
| **Image of Better Response** |  |
| **Description of Weaker Responses** | Weaker responses showed that candidates struggled to plot points on the graph using the values of x and y coordinates. They also faced difficulty in understanding the concept of straight lines which impacted their performance in the question. |
| **Image of Weaker Response** |  |

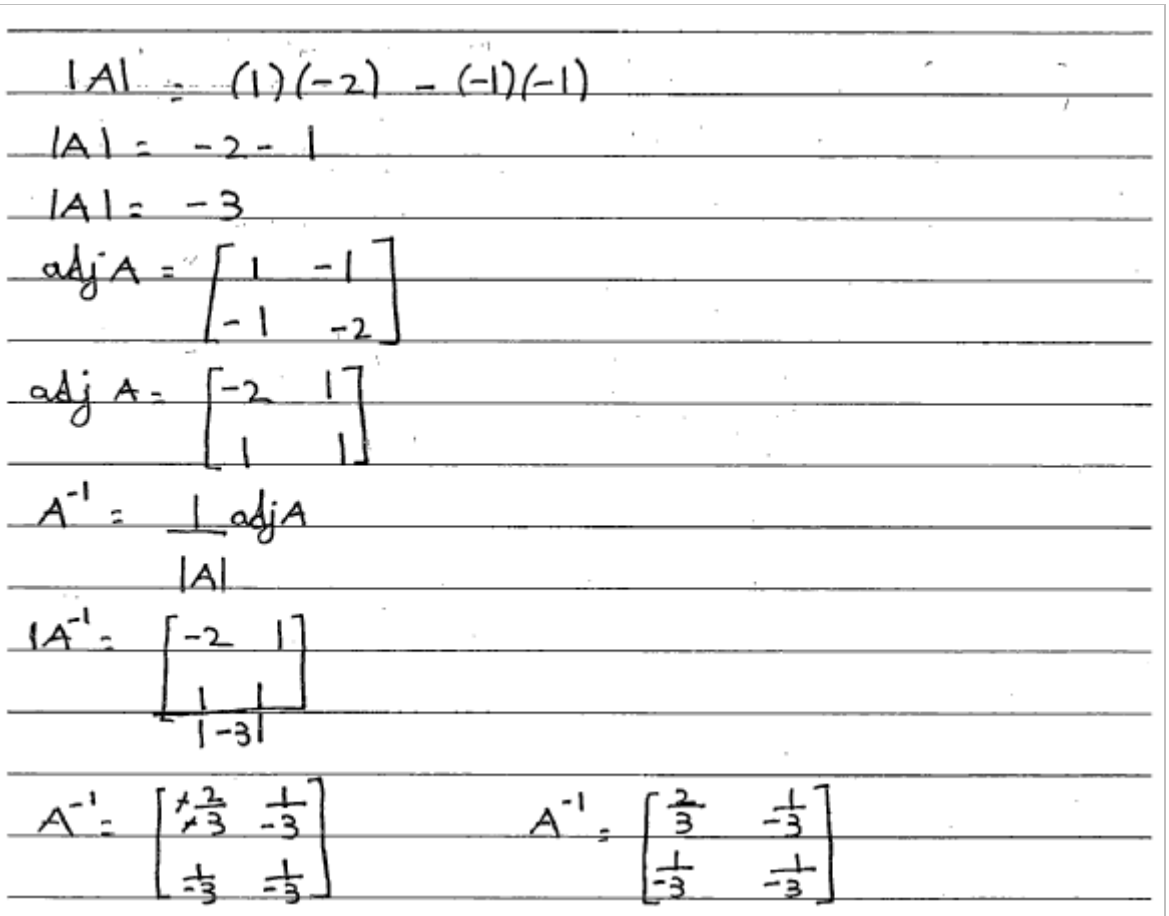
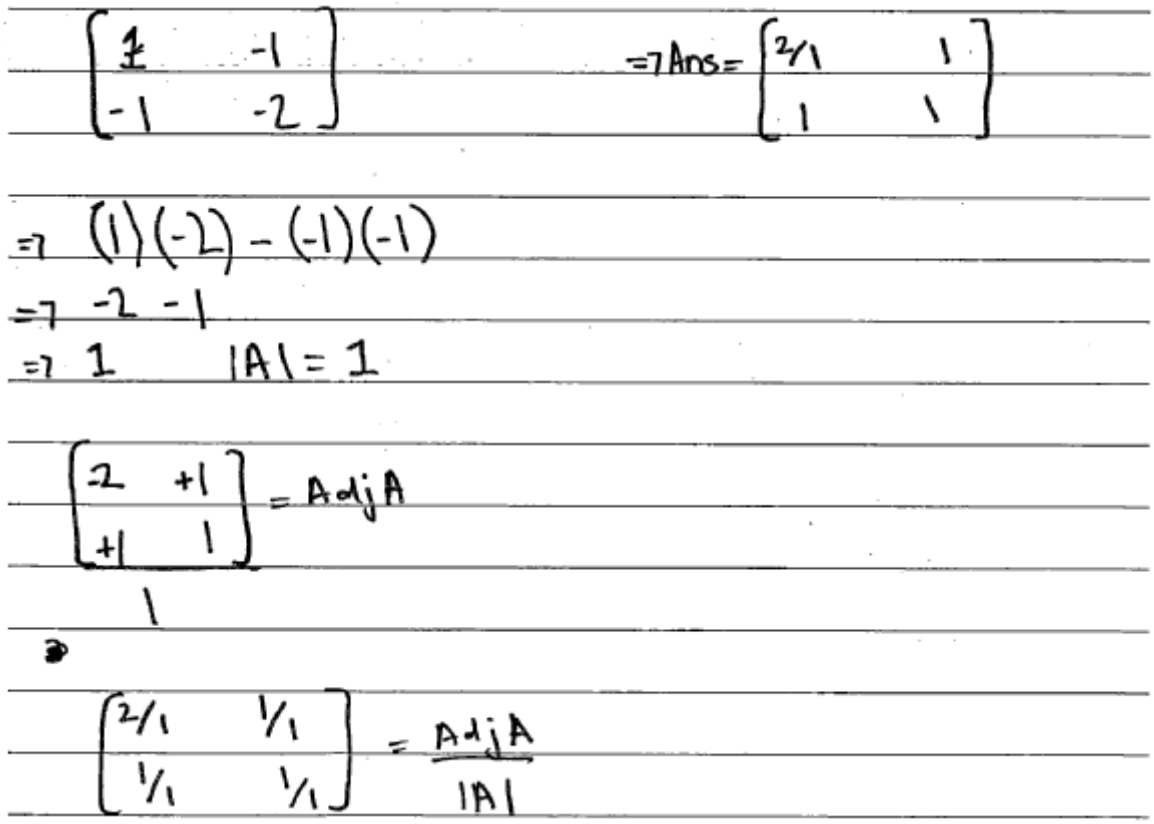
**Suggestions for improvement (Highlight all that apply)**

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| **Any Additional Suggestion:** | | |

**Question No. 8a**

Candidates were given the choice to attempt any ONE out of the two questions: 8a and 8b. Majority of the candidates attempted 8b.

| | |
|---|---|
| **Question Text** | Use the adjoint method to find the inverse of the matrix $A=\begin{bmatrix} 1 & -1 \\ -1 & -2 \end{bmatrix}$. |
| **SLO No.** | 9.5.8 |
| **SLO Text** | Apply adjoint method to calculate the inverse of a matrix. |
| **Max Marks** | 3 |
| **Cognitive Level** | A |
| **Checking Hints** | 1 mark for correct calculation of determinant<br>1 mark for correct calculation of adjoint<br>1 mark for substituting in the formula for inverse matrix and its calculation |
| **Overall Performance** | This question involved the inverse method, where candidates required to apply three concepts: determinant, adjoint and then inverse. Many candidates demonstrated their proficiency by executing each step clearly and methodically. |
| **Description of Better Responses** | Better responses demonstrated that candidates were proficient in calculating both the determinant and the adjoint of the matrix, which facilitated a smooth process for finding the matrix's inverse. They accurately performed multiplication and addition of the matrix elements, ensuring all calculations were precise. Furthermore, they skillfully determined the inverse of the matrix by correctly applying the appropriate signs in their solutions and effectively cancelling them out, leading to a flawless final result. |

| | |
|---|---|
| **Image of Better Response** |  |
| **Description of Weaker Responses** | Despite being considered a familiar question, candidates made significant mistakes that revealed misconceptions related to the concept of Matrices. Common errors were making sign mistakes when calculating the determinant and calculation errors while performing simplification in the method. |
| **Image of Weaker Response** |  |

**Suggestions for improvement (Highlight all that apply)**

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| **Any Additional Suggestion:** | | |

| Question No. 8b | |
|---|---|
| **Question Text** | Solve the matrix equation using Crammer's rule to find the value of $x$.<br>$\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ |
| **SLO No.** | 9.6.2 |
| **SLO Text** | Solve a system of two linear equations in two variables using ii) Crammer's rule |
| **Max Marks** | 3 |
| **Cognitive Level** | A |
| **Checking Hints** | 1 mark for the determinant of the coefficient matrix<br>1 mark for substituting the solution matrix in the first column<br>1 mark for calculating as $x = -\frac{2}{5}$ (this mark would be given, even if the above calculation is incorrect) |
| **Overall Performance** | Most of the candidates attempted this question. Overall, many candidates found Cramer's rule easy and straightforward because it primarily relied on calculating the determinant of a matrix. |
| **Description of Better Responses** | Better responses demonstrated a strong understanding of both the coefficient matrix and the variable matrix in Cramer's rule. They effectively applied the correct formula to calculate the value of 'x', showcasing their proficiency in solving problems using this method. |

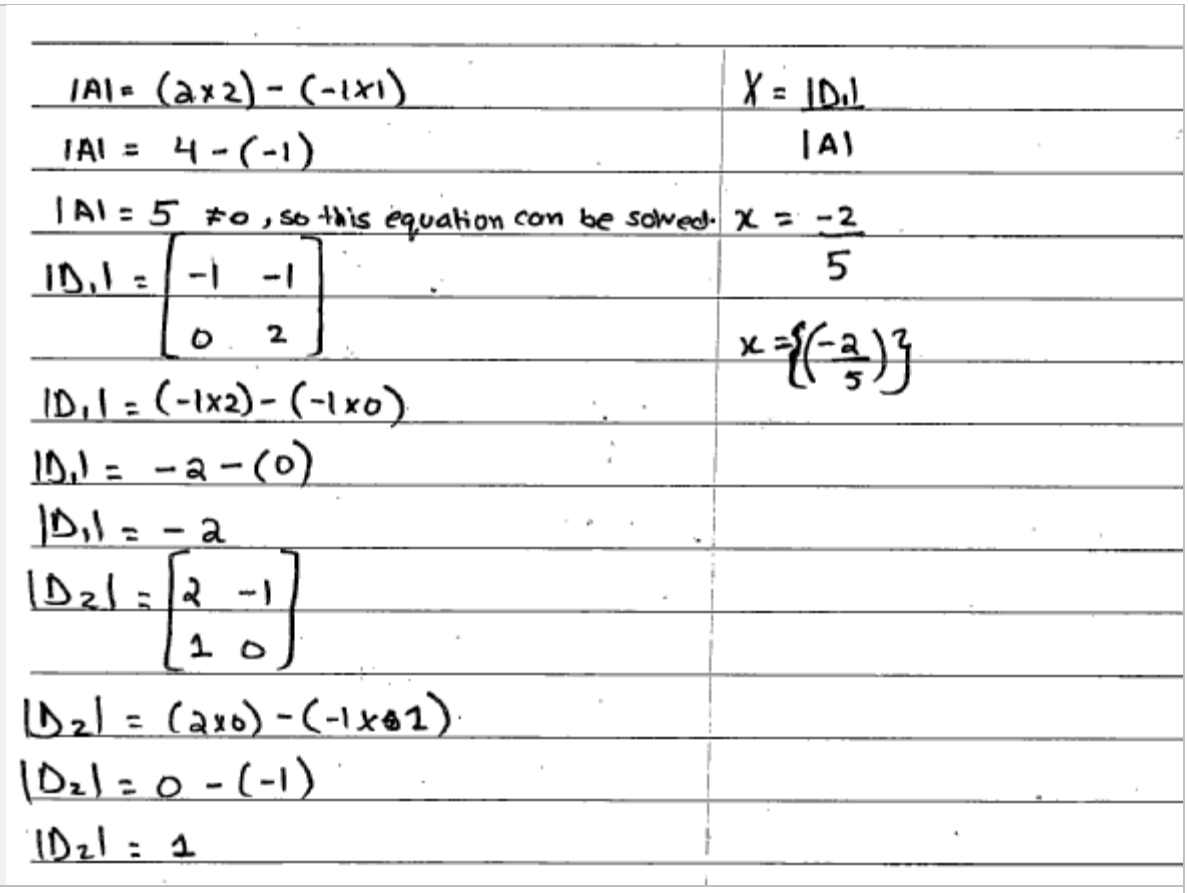
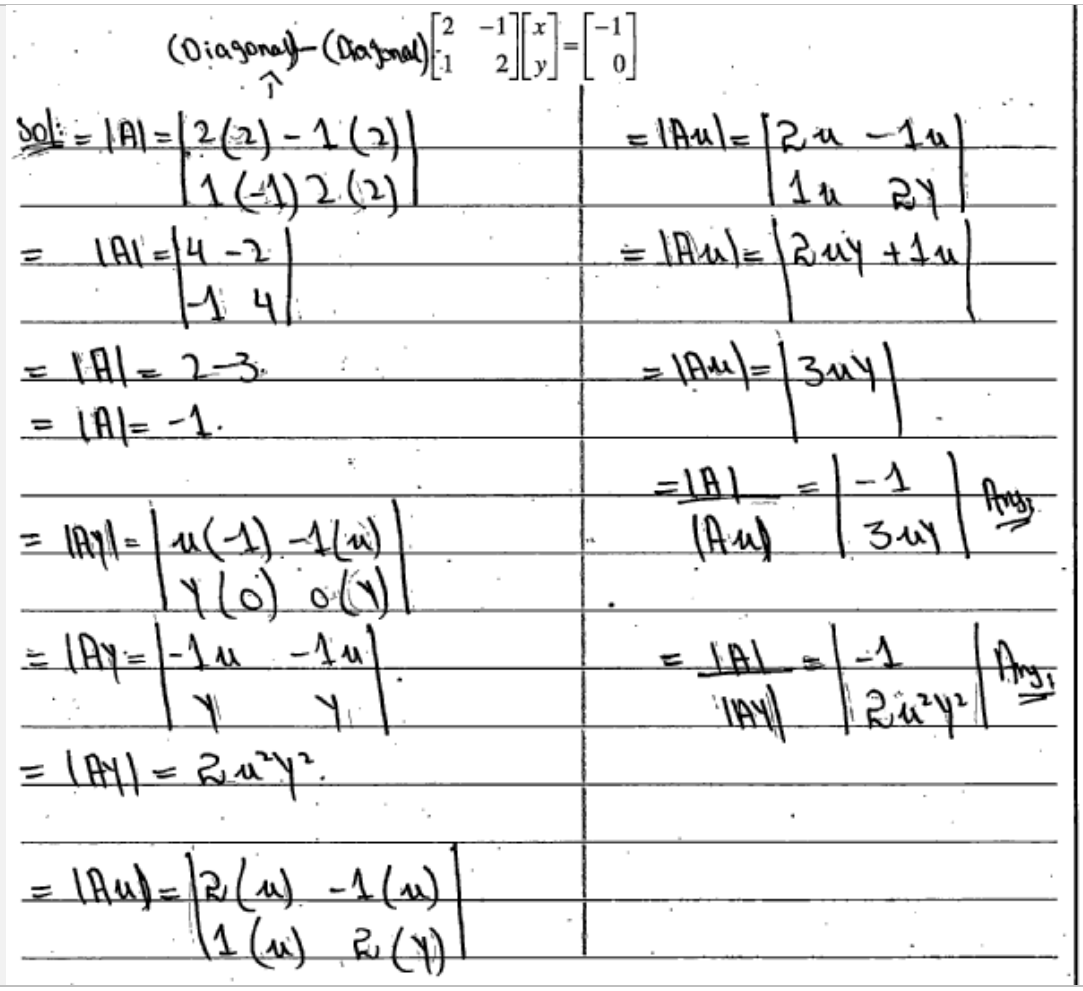
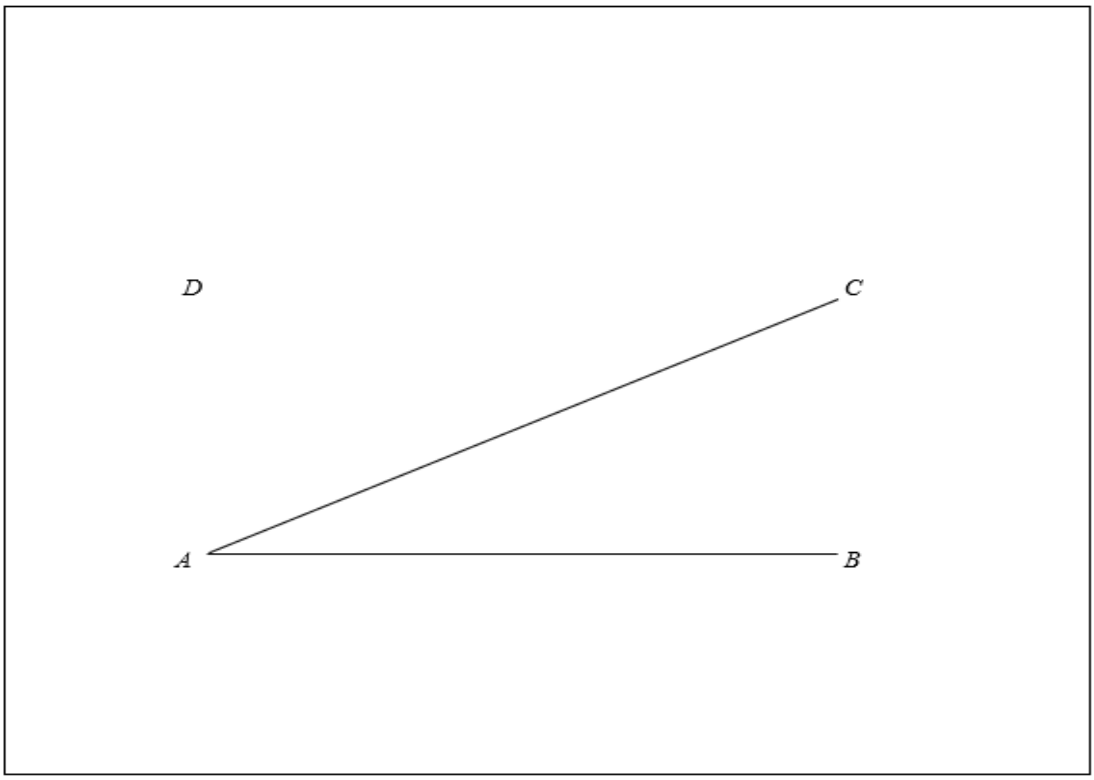
| | |
|---|---|
| **Image of Better Response** |  |
| **Description of Weaker Responses** | Weaker responses did not fully grasp the question's requirements, leading to applying the Inverse method instead of Cramer's rule. Secondly, despite their knowledge of Cramer's rule, some candidates made mistakes during the simplification process. These errors often stemmed from misunderstandings regarding coefficients, variables or sign conventions, resulting in deductions in their scores. |

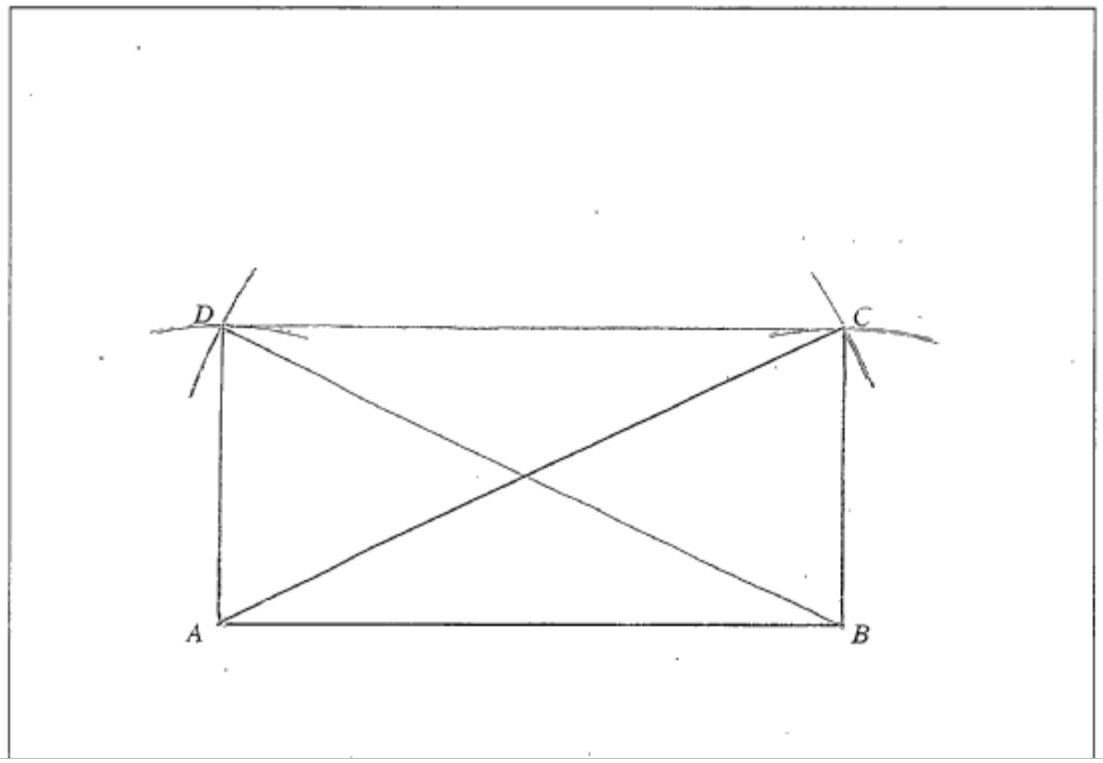
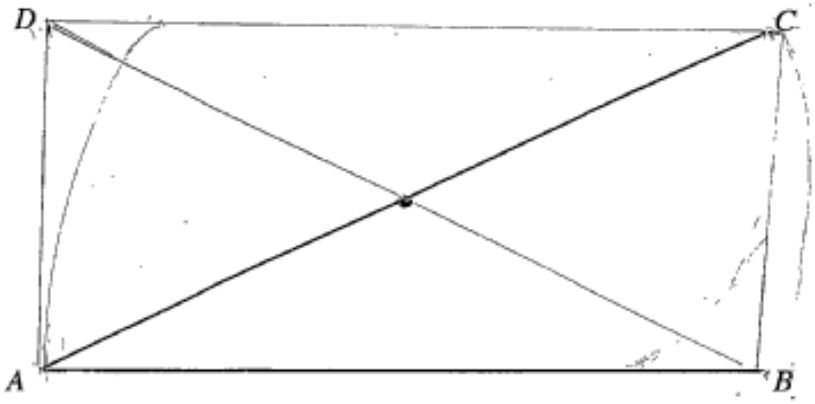
**Image of Weaker Response**




**Suggestions for improvement (Highlight all that apply)**

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| **Any Additional Suggestion:** | | |

| Question No. 9 | |
|---|---|
| **Question Text** | Q.9. (T<br><br>The given diagram shows the diagonal *AC* and the side *AB* of a rectangle. With the help of *AC* and *AB*, draw the rectangle using compass.<br><br>  <br> |
| **SLO No.** | 10.2.1 |
| **SLO Text** | Draw a rectangle when: b. diagonal and one side are given. |
| **Max Marks** | 3 |
| **Cognitive Level** | A |
| **Checking Hints** | 1 mark for drawing the side *BC*, showing the arcs<br>1 mark for drawing the line BE, hence drawing the side *BC*<br>1 mark for connecting *C* to *D* using arcs |
| **Overall Performance** | This question required candidates to draw a rectangle with the given information, using a compass. A good number of candidates got full marks by meeting the demand of the question. |
| **Description of Better Responses** | Better responses successfully constructed a rectangle using compass by meticulously fulfilling all the requirements. They accurately created arcs and joined them carefully to form the sides of the rectangle, demonstrating precision and skill in geometric construction. |

| | |
|---|---|
| **Image of Better Response** |  |
| **Description of Weaker Responses** | In this type of question, constructing the correct geometric figures demands careful and methodical work. Students failed to use the correct arcs to accurately draw a rectangle, resulting in a lack of precision and accuracy in their solutions. |
| **Image of Weaker Response** |  |

**Suggestions for improvement (Highlight all that apply)**

| Maximising SLO Achievement | Preferred Pedagogy Used for this SLO | Assessment Strategies |
|---|---|---|
| • Identify the expectation of command words (use Command Word Guide)<br>• Ensure the content is taught at the relevant cognitive level<br>• Identify necessary content | • Story Board<br>• Cause and Effect<br>• Fish and Bone<br>• Concept Mapping<br>• Audio Visual Resources<br>• Think, Pair and | • Past paper questions<br>• Discussion on E-Marking Notes<br>• AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login |

| | | |
|---|---|---|
| required (skills + concepts)<br>• Review past paper questions on the concept<br>• Utilise the resource guide for additional materials | Share<br>• Knowledge Platform videos<br>• Questioning Technique (Socratic approach)<br>• Practical Demonstration |  |
| **Any Additional Suggestion:** | | |

**Annexure A: Pedagogies Used for Teaching the SLOs**

**Pedagogy:** Storyboard
**Description:** A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.

**Example:** In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

**Pedagogy:** Cause and Effect
**Description**: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.

**Example:** In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

**Pedagogy:** Fish and Bone
**Description:** A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.

**Example:** During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

**Pedagogy:** Concept Mapping
**Description:** An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.

**Example:** In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

**Pedagogy:** Audio Visual Resources

**Description:** Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.

**Example:** In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

**Pedagogy:** Think, Pair, and Share

**Description:** A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.

**Example:** In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

**Pedagogy:** Questioning Technique (Socratic Approach)

**Description:** Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.

**Example:** In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

**Pedagogy:** Practical Demonstration

**Description:** A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.

**Example:** In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

**Acknowledgements**

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Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.

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